

4. Scientific method

a method of procedure consisting of systematic observation, measurement, and experiment, and the formulation, testing, and modification of hypotheses.

What is "**Pseudoscience**?"

What is the scientific assessment of
eyewitness testimony as a form of
evidence?

A **pseudoscience** is a fake science that makes claims based on faulty or nonexistent scientific evidence. In most cases, these pseudosciences present claims in a way that makes them seem possible, but with little or no **empirical** support for these claims.

Graphology, numerology, and astrology, are all examples of pseudosciences. In many cases, these pseudosciences rely on **anecdotes** and testimonials to back up their often **outlandish** claims.

How to Identify Science vs. Pseudoscience

If you are trying to determine if something is a pseudoscience, there are a few key things you can look for:

Consider the purpose. Science is focused on helping people develop a deeper, richer and fuller understanding of the world. Pseudoscience often focuses on furthering some type of ideological agenda.

Consider how challenges are dealt with. Science welcomes challenges and attempts to disprove or refute different ideas. Pseudoscience, on the other hand, tends to greet any challenges to its dogma with hostility.

Look at the research. Science is supported by a deep and ever-growing body of knowledge and research. Ideas around the topic may have changed over time as new things are discovered and new research is performed. Pseudoscience tends to be fairly static. Little may have changed since the idea was first introduced and new research might not exist.

Can it be proven false? Falsifiability is a key hallmark of science. This means that if something is false, researchers could prove that it was false. Many pseudoscientific claims are simply untestable, so there is no way for researchers to prove these claims false.

Example

Phrenology is a good example of how a pseudoscience can capture the public attention and become popular. According to the ideas behind phrenology, bumps on the head were thought to reveal aspects of an individual's personality and character. Physician Franz Gall first introduced the idea during the late 1700s and suggested that the bumps on a person's head corresponded to the physical features of the brain's cortex.

Gall studied the skulls of individuals in hospitals, prisons, and asylums and developed a system of diagnosing different characteristics based upon the bumps of a person's skull. His system included 27 "faculties" that he believed directly corresponded to certain parts of the head.

Like other pseudosciences, Gall's research methods lacked scientific rigor. Not only that, any contradictions to his claims were simply ignored. Gall's ideas outlived him and grew wildly popular during the 1800s and 1900s, often as a form of popular entertainment. There were even phrenology machines that would be placed over a person's head. Spring-loaded probes would then provide a measurement of different parts of the skull and calculate the individual's characteristics.

While phrenology was eventually dismissed as a pseudoscience, it did have an important influence on the development of modern neurology. Gall's idea that certain capabilities were linked to certain parts of the brain led to a growing interest in the idea brain localization, or the notion that certain functions were linked to specific areas of the brain. Further research and observations helped researchers gain a greater understanding of how the brain is organized and the functions of different areas of the brain.

Sources:

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Megendie, F. (1855). *An elementary treatise on human physiology*. Harper and Brothers.

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Wixted, J. (2002). Methodology in experimental psychology. Capstone.

STEPS of the SCIENTIFIC METHOD

- **STEP 1: ASK a QUESTION:**

The scientific method starts when you ask a question about something that you observe:

How, What, When, Who, Which, Why, or Where?

- And, in order for the Scientific Method to answer the question, it must be about something that you can measure, preferably with a number.

- **STEP 2: DO BACKGROUND RESEARCH:**

Rather than starting from scratch in putting together a plan for answering your question, you want to be a savvy scientist using library and Internet research to help you find the best way to do things and ensure that you don't repeat mistakes from the past.

- **STEP 3: CONSTRUCT a HYPOTHESIS:**

A hypothesis is an educated guess about how things work:

“if _____ (*I do this*) _____, then _____ (*this*) will happen.”

You must state your hypothesis in a way that you can easily measure, and of course, your hypothesis should be constructed in a way to help you answer your original question.

- **STEP 4: TEST YOUR HYPOTHESIS
BY DOING AN EXPERIMENT:**

Your experiment tests whether your hypothesis is true or false. It is important for your experiment to be a fair test. You conduct a fair test by making sure that you change only one factor at a time while keeping all other conditions the same.

You should also repeat your experiments several times to make sure that the first results weren't just an accident.

- **STEP 5: ANALYZE YOUR DATA AND DRAW A CONCLUSION:**

Once your experiment is complete, you collect your measurements and analyze them to see if your hypothesis is true or false.

Scientists often find that their hypothesis was false, and in such cases they will construct a new hypothesis starting the entire process of the scientific method over again. Even if they find that their hypothesis was true, they may want to test it again in a new way.

- **STEP 6: COMMUNICATE YOUR RESULTS:**

Lastly, communicate your results to others in a final report. Professional scientists do this by publishing their final report in a scientific journal or by presenting their results on a poster at a scientific meeting.

Scientific Method Reviewed

- ASK Question
- Do Background Research
 - Construct Hypothesis
- Test with an Experiment
- Analyze Results and Draw Conclusions
 - Hypothesis is True
 - If: Hypothesis is False (Go back to Step 3)
 - Report Results

SELECT A SCIENTIFIC RESEARCH TOPIC

HOW TO SELECT A SCIENTIFIC RESEARCH TOPIC

Step 1: Brainstorm for ideas

Step 2: Read General Background Information

Step 3: Focus on Your Topic

Step 4: Make a List of Useful Keywords

Step 5: Be Flexible

Step 6: Define Your Topic as a Focused Research Question

Step 7: Research and Read More About Your Topic

Step 8: Formulate an article title and sub-titles